

AI Study Engine - Complete Technology Stack Guide

1. Overall Architecture

Frontend: React + Tailwind CSS + React Router Backend: FastAPI (Python) Database: MongoDB
Atlas Authentication: JWT + Google OAuth AI Layer: Gemini API File Storage: Cloudinary /
Firebase Storage Deployment: Vercel + Render Version Control: Git + GitHub

2. Dashboard Module

Frontend: • React • Tailwind CSS • ShadCN UI • Recharts Backend: • FastAPI Database: •
MongoDB Purpose: Display statistics, recommendations, tasks and progress analytics.

3. AI Mentor Module

Frontend: • React Chat Interface • Tailwind CSS Backend: • FastAPI AI: • Gemini API • LangChain
Database: • MongoDB Chat History Features: Career Coach Study Coach Resume Coach
Interview Coach Productivity Coach

4. Knowledge Vault Module

Frontend: • React Upload Components • Tailwind Backend: • FastAPI Upload APIs Storage: •
Cloudinary or • Firebase Storage Database: • MongoDB Metadata AI: • Gemini API • PyPDF
Features: Summaries Keywords Flashcards MCQs

5. Roadmaps & Guides

Frontend: • React • Progress Components Backend: • FastAPI AI: • Gemini API Database: •
MongoDB Purpose: Generate learning roadmaps and structured guides.

6. Internship Recommendation Engine

Frontend: • Internship Cards • Filters • Search Backend: • FastAPI Machine Learning: • Scikit Learn
• TF-IDF Vectorizer • Cosine Similarity Database: • MongoDB Purpose: Match student profiles with
internships.

7. Resume Studio

Frontend: • Resume Upload UI Backend: • FastAPI Libraries: • pdfplumber • PyPDF2 AI: • Gemini
API Database: • MongoDB Features: ATS Score Resume Analysis Project Suggestions

8. Study Assistant

Frontend: • PDF Upload Interface Backend: • FastAPI AI: • Gemini API Libraries: • PyPDF2 •
pdfplumber Outputs: MCQs Quizzes Flashcards Summaries

9. Productivity Hub

Frontend: • Calendar Components • Task Components Backend: • FastAPI Database: • MongoDB
Features: Tasks Goals Habits Deadlines

10. Opportunity Center

Frontend: • Cards • Search Backend: • FastAPI Data Sources: • APIs • Custom Dataset Database: • MongoDB Categories: Hackathons Scholarships Internships Contests

11. Analytics Module

Frontend: • Recharts Backend: • FastAPI Database: • MongoDB Metrics: Study Hours Tasks Completed Applications Roadmap Progress

12. Authentication System

Frontend: • React Forms Backend: • FastAPI Authentication: • JWT • Google OAuth Database: • MongoDB Features: Signup Login Protected Routes

13. Database Collections

Users Documents Roadmaps Tasks ChatHistory Internships Applications Opportunities Analytics

14. MongoDB Schema

Users: name email branch skills goal Documents: title summary tags fileUrl Tasks: title deadline status

15. API Structure

POST /login POST /register POST /upload POST /chat POST /resume-analysis POST /roadmap GET /internships GET /opportunities GET /analytics

16. Folder Structure

frontend/ ■ pages ■ components ■ services ■ hooks ■ assets backend/ ■ api ■ models ■ services ■ ai ■ database ■ utils

17. Deployment

Frontend: Vercel Backend: Render Database: MongoDB Atlas Storage: Cloudinary Domain: Namecheap / GoDaddy

18. Team Responsibilities

Frontend Developer: Dashboard Vault Roadmaps Frontend Developer: AI Mentor Resume Studio Backend Developer: API Database Authentication AI Developer: Gemini Recommendations Prompt Engineering

19. Hackathon MVP Stack

React Tailwind FastAPI MongoDB Atlas Gemini API Cloudinary This is enough to build the MVP.

20. Future Advanced Stack

Vector Database: Pinecone RAG: LangChain Notifications: Firebase Mobile App: React Native AI Voice Assistant: Whisper + Gemini